

Project Title: Network Intrusion Detection System using ML

Problem Statement (AICTE PS 40):

Create a robust network intrusion detection system (NIDS) using machine learning. The system should be capable of analyzing network traffic data to identify and classify various types of cyber-attacks (e.g., DoS, Probe, R2L, U2R) and distinguish them from normal network activity. The goal is to build a model that can effectively secure communication networks by providing an early warning of malicious activities.

Proposed Solution:

- Proposed an AI-powered Network Intrusion Detection System built using **IBM watsonx.ai Studio** and **AutoAI**.
- Utilized a **Kaggle NSL-KDD** network traffic connection dataset featuring key network metrics (e.g., protocol types, connection durations, failed logins, and byte metrics).
- Categorized columns and grouped attack labels into **5 specific classification profiles**: *Normal, DoS, Probe, R2L, and U2R*.
- AutoAI automatically performed data preprocessing, feature engineering, algorithm selection, and hyperparameter optimization across multiple pipelines.
- The best-performing pipeline was selected, deployed as a live scoring API on IBM Cloud, and connected to an interactive Streamlit dashboard for real-time predictions.

Technology Used:

- **IBM Cloud** - provides the infrastructure to build, deploy, and scale the intrusion detection model efficiently. IBM Cloud Lite enables a reliable, secure, and scalable environment for hosting the deployed model.
- **IBM watsonx.ai AutoAI** - automated model training and selection.
- **IBM Bob** - AI coding agent used to build the data pipeline and Streamlit dashboard.
- **ML Algorithm/Model Used:** *Snap Decision Tree Classifier*
- **Accuracy Obtained:** *0.997*
- **Enhancements and Hyperparameter Tuning:** *HOP-1*

Expected Outcome:

A deployed, functioning machine learning system that can classify live network traffic input into normal activity or a specific attack category with high accuracy, demonstrating a practical, end-to-end application of AI in cybersecurity from raw data to a real-time, publicly accessible prediction tool.